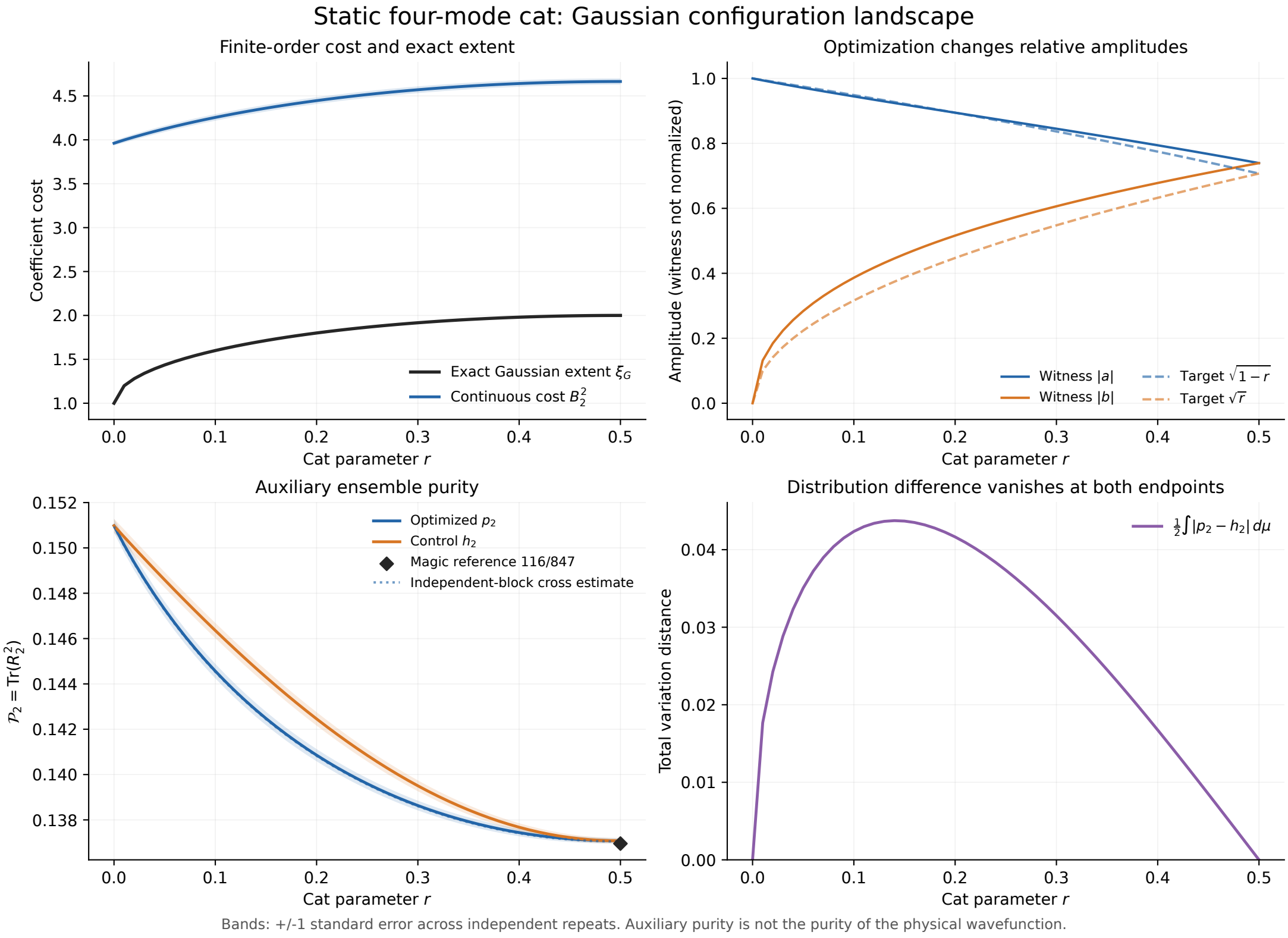
Static four-mode cat: Gaussian configuration landscape
Finite-order cost and exact extent
Coefficient cost
Cat parameter r
Exact Gaussian extent ξ_G
Continuous cost B_2^2
Optimization changes relative amplitudes
Amplitude (witness not normalized)
Witness $|a|$
Witness $|b|$
Target $\sqrt{1-r}$
Target $\sqrt{r}$
Auxiliary ensemble purity
$\mathcal{P}_2 = \mathrm{Tr}(R_2^2)$
Optimized p_2
Control h_2
Magic reference 116/847
Independent-block cross estimate
Distribution difference vanishes at both endpoints
Total variation distance
$\frac{1}{2}\int |p_2 - h_2|\, d\mu$
Bands: +/-1 standard error across independent repeats. Auxiliary purity is not the purity of the physical wavefunction.